

**SAFETY DATA SHEET**

Creation Date 01-Sep-2009

Revision Date 25-Mar-2024

Revision Number 3

**1. Identification**

**Product Name** Isopropanol

**Cat No. :** C38392

**CAS No** 67-63-0  
**Synonyms** 2-Propanol; IPA; Isopropyl alcohol; Propan-2-ol; Isopropanol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

**2. Hazard(s) identification****Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                                   |            |
|-------------------------------------------------------------------|------------|
| Flammable liquids                                                 | Category 2 |
| Serious Eye Damage/Eye Irritation                                 | Category 2 |
| Specific target organ toxicity (single exposure)                  | Category 3 |
| Target Organs - Respiratory system, Central nervous system (CNS). |            |
| Specific target organ toxicity - (repeated exposure)              | Category 2 |
| Target Organs - Kidney, Liver.                                    |            |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Highly flammable liquid and vapor  
Causes serious eye irritation

May cause respiratory irritation  
May cause drowsiness or dizziness  
May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Wear protective gloves/protective clothing/eye protection/face protection  
Keep cool

**Response**

Get medical attention/advice if you feel unwell

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component         | CAS No  | Weight % |
|-------------------|---------|----------|
| Isopropyl alcohol | 67-63-0 | >95      |

### 4. First-aid measures

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if

|                                            |                                                                                                                                                                                           |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | symptoms occur.                                                                                                                                                                           |
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                                                                                                |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Get medical attention.                                                                                                                                            |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. May cause central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                     |

## 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | Water may be ineffective                                                                                            |
| <b>Flash Point</b>                      | 12 °C / 53.6 °F                                                                                                     |
| <b>Method -</b>                         | Abel Closed Cup (BS 2000 Part 170, IP 170, AS/NZS 2106)                                                             |
| <b>Autoignition Temperature</b>         | 425 °C / 797 °F                                                                                                     |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | 12 vol %                                                                                                            |
| <b>Lower</b>                            | 2 vol %                                                                                                             |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). peroxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 3                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                                                            |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Avoid contact with skin, eyes or clothing.                                                                                                           |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                                                                                                                                         |
| <b>Methods for Containment and Clean Up</b> | Prevent further leakage or spillage if safe to do so. Remove all sources of ignition. Soak up with inert absorbent material. Take precautionary measures against static discharges. Use spark-proof tools and explosion-proof equipment. Keep in suitable, closed containers for disposal. |

## 7. Handling and storage

|                 |                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Keep away from open flames, hot surfaces and sources of ignition. Use spark-proof tools and explosion-proof equipment. Use |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

only non-sparking tools. Take precautionary measures against static discharges. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

**Storage.**

Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Acids. Halogens. Acid anhydrides.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component         | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          | NIOSH                                                                                                         | Mexico OEL (TWA)              |
|-------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------|
| Isopropyl alcohol | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1225 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 400 ppm |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

**Recommended Filter type:**

Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                                                         |
|----------------------------------|---------------------------------------------------------|
| Physical State                   | Liquid                                                  |
| Appearance                       | Colorless                                               |
| Odor                             | Alcohol-like                                            |
| Odor Threshold                   | No information available                                |
| pH                               | 7 1% aq. sol                                            |
| Melting Point/Range              | -89.5 °C / -129.1 °F                                    |
| Boiling Point/Range              | 81 - 83 °C / 177.8 - 181.4 °F @ 760 mmHg                |
| Flash Point                      | 12 °C / 53.6 °F                                         |
| Method -                         | Abel Closed Cup (BS 2000 Part 170, IP 170, AS/NZS 2106) |
| Evaporation Rate                 | 1.7                                                     |
| Flammability (solid,gas)         | Not applicable                                          |
| Flammability or explosive limits |                                                         |
| Upper                            | 12 vol %                                                |

|                                        |                                                       |
|----------------------------------------|-------------------------------------------------------|
| Lower                                  | 2 vol %                                               |
| Vapor Pressure                         | 43 mmHg @ 20 °C                                       |
| Vapor Density                          | 2.1 @ 20 °C / 68 °F                                   |
| Specific Gravity                       | 0.785                                                 |
| Solubility                             | Miscible with water                                   |
| Partition coefficient; n-octanol/water | No data available                                     |
| Autoignition Temperature               | 425 °C / 797 °F                                       |
| Decomposition Temperature              | No information available                              |
| Viscosity                              | 2.27 mPa.s at 20 °C                                   |
| Molecular Formula                      | C3 H8 O                                               |
| Molecular Weight                       | 60.1                                                  |
| VOC Content(%)                         | 100% (Organic Carbon (by mass) = 59.9 %) (EC/1999/13) |
| Refractive index                       | 1.377 at 20 °C / 68 °F (ASTM D-1218)                  |
| Surface tension                        | 22.7 mN/m at 20 °C / 68 °F                            |
| Coefficient of expansion               | 0.0009 / °C                                           |
| Dielectric constant                    | 18.6 at 20 °C / 68 °F                                 |
| Heat of vapourisation                  | 665 J/g                                               |
| Specific heat capacity                 | 3 kJ/kg °C at 20 °C / 68 °F                           |
| Thermal conductivity                   | 0.137 W/m °C at 20 °C / 68 °F                         |

## 10. Stability and reactivity

|                                  |                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                 |
| Stability                        | Stable under normal conditions.                                                            |
| Conditions to Avoid              | Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Acids, Halogens, Acid anhydrides                                  |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), peroxides                         |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                   |
| Hazardous Reactions              | None under normal processing.                                                              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component         | LD50 Oral                                  | LD50 Dermal         | LC50 Inhalation       |
|-------------------|--------------------------------------------|---------------------|-----------------------|
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat ) | 72.6 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes and skin                                                              |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component         | CAS No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------|---------|------------|------------|------------|------------|------------|
| Isopropyl alcohol | 67-63-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

|                                                                  |                                                                                                                                                                  |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developmental Effects</b>                                     | No information available.                                                                                                                                        |
| <b>Teratogenicity</b>                                            | No information available.                                                                                                                                        |
| <b>STOT - single exposure</b><br><b>STOT - repeated exposure</b> | Respiratory system Central nervous system (CNS)<br>Kidney Liver                                                                                                  |
| <b>Aspiration hazard</b>                                         | No information available                                                                                                                                         |
| <b>Symptoms / effects, both acute and delayed</b>                | May cause central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>                           | No information available                                                                                                                                         |
| <b>Other Adverse Effects</b>                                     | The toxicological properties have not been fully investigated.                                                                                                   |

## 12. Ecological information

### Ecotoxicity

. Do not empty into drains.

| Component         | Freshwater Algae                                                                                           | Freshwater Fish                                                                                                                                                                                                          | Microtox                                              | Water Flea                                      |
|-------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------|
| Isopropyl alcohol | EC50: > 1000 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h<br>(Desmodesmus subspicatus) | LC50: = 9640 mg/L, 96h<br>flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h<br>(Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h<br>static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h<br>(Daphnia) | = 35390 mg/L EC50<br>Photobacterium phosphoreum 5 min | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

| Component         | log Pow |
|-------------------|---------|
| Isopropyl alcohol | 0.05    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                      |             |
|----------------------|-------------|
| UN-No                | UN1219      |
| Proper Shipping Name | Isopropanol |
| Hazard Class         | 3           |
| Packing Group        | II          |

### TDG

|                      |             |
|----------------------|-------------|
| UN-No                | UN1219      |
| Proper Shipping Name | ISOPROPANOL |
| Hazard Class         | 3           |
| Packing Group        | II          |

### IATA

|       |        |
|-------|--------|
| UN-No | UN1219 |
|-------|--------|

**Proper Shipping Name** Isopropanol  
**Hazard Class** 3  
**Packing Group** II  
**IMDG/IMO**  
**UN-No** UN1219  
**Proper Shipping Name** Isopropanol (Isopropyl alcohol)  
**Hazard Class** 3  
**Packing Group** II

## 15. Regulatory information

### United States of America Inventory

| Component         | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------|---------|------|-----------------------------------------------|-----------------------------|
| Isopropyl alcohol | 67-63-0 | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)** Not applicable

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component         | CAS No  | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-------------------|---------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Isopropyl alcohol | 67-63-0 | X   | -   | 200-661-7 | X     | X    | X    | X    | X     | KE-29363 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component         | CAS No  | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-------------------|---------|----------|-------------------------------|---------------------------------|
| Isopropyl alcohol | 67-63-0 | >95      | 1.0 %                         | -                               |

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA** - Occupational Safety and Health Administration Not applicable

#### CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and

Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------|---------------|------------|--------------|----------|--------------|
| Isopropyl alcohol | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

Serious risk, Grade 3

**Authorisation/Restrictions according to EU REACH**

| Component         | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol | 67-63-0 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component         | CAS No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Isopropyl alcohol | 67-63-0 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component         | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Isopropyl alcohol | 67-63-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |



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**16. Other information**

|                         |                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department<br>Email: chem.techinfo@thermofisher.com<br>www.thermofisher.com |
| <b>Creation Date</b>    | 01-Sep-2009                                                                                                  |
| <b>Revision Date</b>    | 25-Mar-2024                                                                                                  |
| <b>Print Date</b>       | 25-Mar-2024                                                                                                  |
| <b>Revision Summary</b> | New emergency telephone response service provider.                                                           |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**